CLASS 07

Experimental Design in Political Science

GOVT 10: Quantitative Political Analysis

This chapter is about designing experiments of your own. It starts with the two decisions at the heart of any experiment — what to vary and what to measure — then surveys the main design families (vignette, conjoint, economic game, factorial, and field) and the implementation threats (weak manipulations, demand effects, inattentive respondents, non-compliance, and differential attrition) that separate a credible experiment from a misleading one.

1 Experimental Design in Political Science

Earlier in the course we learned how to summarize and visualize data that already exists. But political scientists are not content to observe. We want to know whether a campaign ad actually changes attitudes, whether sending a mailer actually increases turnout, whether transparency actually reduces corruption. These are causal questions, and the cleanest tool we have for answering them is the experiment.

The logic of an experiment is simple. We take a group of people, randomly split them into conditions, expose each condition to a different treatment, and compare what happens. Because the split was random, the only systematic difference between the groups is the treatment we applied. Any difference in outcomes can be attributed to the treatment rather than to preexisting differences between participants.

This logic is so powerful that experiments have come to dominate large parts of political science. But the basic idea hides a lot of design choices. What is the treatment going to look like? Will participants see one condition or several? How many factors will we manipulate at once? What does the control group experience? Before surveying the main families of experimental designs, it helps to start with the two decisions that define every experiment.

1.1 The Two Big Decisions

Strip an experiment down to its core and it is two choices: what you **vary** and what you **measure**. The thing you vary is the treatment, or independent variable — call it X. The thing you measure is the outcome, or dependent variable — call it Y. Random assignment links them: because people are sorted into conditions by chance, any systematic difference in Y can be traced back to X. Most of the craft of experimental design is getting these two choices right, because a clever analysis cannot rescue a muddled treatment or a meaningless outcome.

This is also why experiments are the workhorse of student and thesis research in political science. You usually cannot field a new national survey or assemble an original panel dataset, but you can manufacture a clean comparison: write two versions of a stimulus, randomize which one each participant sees, and measure what happens. With a modest budget and a survey platform, an undergraduate can produce a genuinely causal finding.

1.2 What to Vary: Designing the Treatment

A treatment begins as an abstract idea and has to become something concrete on a screen. You move down a ladder from theory, to construct, to operationalization. Suppose you think *threat* makes people more hawkish. "Threat" is an abstraction; to put it in front of a participant you might describe a hostile country building nuclear weapons and vary whether that country is called a democracy or an autocracy. Tomz and Weeks (2013) did exactly this and found that people are far less willing to strike a democracy — even one building nuclear weapons. The abstract construct became one swappable line of text.

> **Change One Thing at a Time**
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> A good manipulation differs from the control on exactly one dimension. If your "negative ad" differs from the control in tone, script, music, and imagery all at once, a significant effect cannot tell you which feature mattered — this is the **bundle problem**. Hold everything constant except the feature you care about. If you must bundle several features together, you can only claim that the whole package works, not any single part of it.

A treatment also has a strength, or **dose**. If the manipulation is too subtle, you will find no effect and wrongly blame the theory when the real culprit was a weak treatment. Gerber, Green, and Larimer (2008) mailed Michigan households messages with escalating social pressure, from a mild "do your civic duty" to a pointed threat to reveal each household's voting record to its neighbors. The same basic message, delivered at a stronger dose, moved turnout from under two percentage points to more than eight. The lesson for your own work: make the treatment strong enough to matter, and then verify that it landed (see manipulation and attention checks below).

1.3 What to Measure: Choosing the Outcome

Outcomes come in two broad flavors. **Attitudinal** or self-report outcomes — a one-to-seven support scale, a feeling thermometer — are cheap and flexible, but people can posture, guess, or misreport. **Behavioral** outcomes — whether someone donates, signs, clicks, or actually votes — are harder to collect but far more credible, because they record what people did rather than what they said. When you can, measure behavior, or at least an outcome with real consequences. Butler and Broockman (2011), studying whether legislators discriminate against Black constituents, did not ask legislators about their attitudes; they measured whether legislators actually replied to an email.

Two further choices matter. A single survey item is noisy; combining several related items into an index averages out that noise and captures the underlying concept more reliably. And you should fix your one *primary* outcome *before* you see the data. If you measure ten things and report only the two that “worked,” you are not testing a hypothesis, you are fishing — the garden of forking paths. Naming a single primary outcome in advance, with everything else labeled secondary or exploratory, keeps you honest.

> **Timing and Demand Effects**
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> *When* you measure matters as much as what you measure. Many framing and advertising effects are real but fleeting, fading within days. If you measure only immediately after treatment, you learn about a momentary blip, not whether anything sticks; a delayed follow-up tests durability.
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> You also want to avoid **demand effects**. If participants can guess what you are hoping to find, some will play along and others will resist — either way you are measuring their read of your study rather than the treatment. Keep the manipulation and the outcome from sitting too obviously side by side, use neutral wording, and do not telegraph the hypothesis.

1.4 Vignette Experiments

A vignette experiment presents participants with a short hypothetical scenario, then randomly varies one or two details across conditions. The participant reads the scenario, answers some questions about it, and the researcher compares responses across the randomly assigned versions.

Vignettes are popular for three reasons. They let us isolate the effect of a single factor with tight control. They let us study sensitive topics, such as discrimination or political violence, without exposing anyone to actual harm. And they are cheap and fast to run on survey platforms.

> **Example: The Nuclear Taboo**
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> Press, Sagan, and Valentino (2013) wanted to know how strong the norm against nuclear weapons really is. They gave a representative U.S. sample a vignette about an Al Qaeda nuclear lab in Syria, then randomly varied two things: whether the proposed strike was nuclear or conventional, and how effective it would be (90% vs. 70% vs. 45%).
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> The treatment was a single paragraph of text, but it generated five different conditions. Support for using nuclear weapons rose sharply as the conventional alternative became less effective. The ”taboo” turned out to be much weaker than people assumed.

The catch with vignettes is the gap between what people say they would do in a hypothetical and what they actually do in real life. A respondent reading about a fictional candidate is not the same as a voter standing in a booth. Vignettes give us internal validity (we know the treatment caused the response we measured) at some cost to external validity (we are less sure the response generalizes to real political behavior).

Vignettes are also useful for studying things that are hard to manipulate ethically in the real world. We cannot randomly assign respondents to live under different policies, but we can describe two policy scenarios in a paragraph and randomize which one each respondent reads. The trade-off, again, is realism: a paragraph of text is a thin stand-in for actually living under a policy.

1.5 Conjoint Experiments

A vignette manipulates one thing at a time. A conjoint experiment manipulates many things at once. Participants see profiles, often candidate profiles or immigrant profiles, with multiple attributes randomly assigned, and choose between them. By varying everything simultaneously, we can ask which attributes actually drive choice and which ones do not matter as much as people claim.

> A typical immigration conjoint shows participants two hypothetical immigrants side by side. Each immigrant has nine attributes: country of origin, education, profession, language skills, prior trips to the U.S., reason for applying, employment plans, job experience, and gender. Every attribute is randomly assigned. The participant picks which immigrant should be admitted, then sees another pair. After several rounds, the researcher has enough data to estimate the independent effect of each attribute.

Hainmueller and Hopkins (2015) ran exactly this design with 1,407 U.S. adults. The results were surprising. Economic attributes (education, employment plans) dominated decisions. Country of origin had a minimal effect, which challenged the standard story about xenophobia driving immigration attitudes. And the rankings looked remarkably similar across party lines.

Conjoints have two big advantages over vignettes. First, they reduce social desirability bias. When nine things are varying at once, it is hard to guess the “correct” answer, so participants reveal preferences they might otherwise hide. Second, they let you compare attributes against each other. A vignette can tell you that education matters; a conjoint can tell you that education matters more than country of origin.

The trade-offs are real. Conjoints can overwhelm participants if too many attributes are varied. Some randomly generated profiles are implausible (a 25-year-old with a PhD and 20 years of experience). And the choice format pushes participants to compare, which may not be how they would think about the question in real life.

A version of the conjoint that has taken off recently is the candidate-choice design. Participants see pairs of hypothetical candidates with attributes such as party, age, gender, prior office, religion, and policy positions, and pick whom they would vote for. Researchers have used these designs to study how voters trade off candidate identity against ideology, how strong the partisan default is, and whether voters in primaries behave differently than voters in general elections. Because every attribute is randomized, the conjoint can separate the effect of, say, candidate gender from the effect of candidate ideology even though these are correlated in real elections.

1.6 Economic Games

Vignettes and conjoints measure stated preferences. Economic games measure something closer to behavior. Participants are given real money and asked to make decisions that have real consequences. The premise is that putting money on the line reveals what people actually value, not just what they say.

> **Common Games in Political Science**
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> **Dictator game.** One player unilaterally splits money with another. Measures altruism and redistribution preferences.
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> **Ultimatum game.** The proposer offers a split. The responder accepts (both get their share) or rejects (both get nothing). Measures fairness norms.
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> **Trust game.** Player 1 sends some amount, which is tripled, to Player 2. Player 2 decides how much to return. Measures trust and reciprocity.
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> **Public goods game.** Players contribute to a shared pool. Measures cooperation and collective action.

A particularly influential application is Iyengar and Westwood (2015), who ran a trust game with $10 stakes and a 2x2 design crossing partner party (Democrat or Republican) and partner race (White or Black). Partisan discrimination turned out to exceed racial discrimination. People kept more money from co-partisans and less from out-partisans, even when stakes were small.

Games have one big virtue: real stakes. You are observing behavior, not opinions. The downsides are that lab games are simple by design and politics is not, samples tend to skew toward students, and the dollar amounts involved are usually trivial compared to actual political decisions.

A useful way to think about the three designs covered so far is in terms of what they measure. Vignettes measure stated reactions to controlled scenarios. Conjoints measure stated preferences in the face of many simultaneous trade-offs. Games measure behavior under real incentives. None of these is uniformly better; the right choice depends on what you are trying to learn. Political scientists often combine designs (a vignette to establish that an effect exists, a conjoint to identify which features drive it, and a game to verify that the same pattern shows up in behavior) to triangulate on a finding.

1.7 Within-Subjects vs. Between-Subjects

Every experiment has to decide how to assign people to conditions. The two main options are between-subjects and within-subjects assignment.

In a between-subjects design, each participant sees one condition only. We randomly split the sample into a treatment group and a control group, expose each group to its assigned condition, and compare outcomes across people. This is the standard structure for most political experiments.

In a within-subjects design, each participant sees every condition. We measure each person's response under condition A and under condition B, then compare A versus B within each person.

> Suppose you want to test whether a campaign ad shifts attitudes toward a candidate.
>
> **Between-subjects:** 500 people see the ad, 500 see a placebo. Compare the average attitude across groups.
>
> **Within-subjects:** 250 people see the ad first, then the placebo. Another 250 see the placebo first, then the ad. Compare each person's attitude under each condition.
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> The within-subjects design needs roughly half the sample to detect the same effect because each person serves as their own control. The downside is that seeing one condition can change how you respond to the next.

Clifford, Sheagley, and Piston (2021) reanalyzed 12 experiments with nearly 28,000 participants and found that within-subjects designs increased precision by 35 to 40 percent without biasing treatment estimates. Within-subjects works when treatments are conceptually independent, order is randomized, there is enough "washout" between conditions, and learning effects do not carry over. When those conditions fail (when seeing one treatment changes how you respond to the next) between-subjects is safer.

1.8 Factorial Designs

A factorial design varies more than one factor at a time. The simplest version is a 2x2: two factors, each with two levels, producing four conditions.

The key advantage of a factorial design is efficiency. From a single study, you get three tests:

1. The main effect of Factor 1
2. The main effect of Factor 2
3. The interaction between Factors 1 and 2

The interaction is the interesting one. It tells you whether Factor 1's effect depends on Factor 2.

> **Example: Resume Audit**
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> Bertrand and Mullainathan (2004) sent roughly 5,000 fictitious resumes to real job postings in Chicago and Boston. They varied two factors:
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> - **Name signal** (stereotypically White vs. stereotypically Black)
> - **Resume quality** (high vs. low)
>
> White names received about 50% more callbacks than Black names. That is the main effect of name. But the interesting finding was the interaction: improving resume quality helped White applicants more (a 2.3 percentage point gain) than Black applicants (a 0.5 percentage point gain). Higher quality did not close the gap; it widened it.

Factorial designs scale up quickly. A 2x2x2 has eight conditions, a 2x2x2x2 has sixteen. The cost is sample size. To estimate each cell precisely, you need enough participants per cell. Most factorial experiments in political science stick to 2x2 or 2x3 structures for this reason.

Another common factorial application is the credit-claiming literature in American politics. Like Bertrand and Mullainathan, this varies two factors simultaneously. Grimmer, Messing, and Westwood (2012) used a 2x2 design to test whether constituents respond more to message frequency or to dollar amounts. The first factor was frequency (1 message vs. 5 messages from a representative). The second factor was the dollar amount claimed ($15,000 vs. $1.5 million in district spending). Frequency dominated. Sending five small-dollar messages was about 90 times more effective per dollar than a single big-dollar message. Without the factorial structure, the researchers could not have separated these two effects from each other.

1.9 Control Groups and Placebos

A treatment effect is always a comparison. The treatment group sees something; the control group sees something else. If we picked the wrong "something else," we will misinterpret the result.

The cleanest control is a pure control group that sees nothing related to the treatment. But sometimes that is not enough. Suppose a get-out-the-vote experiment compares people who received a mailer to people who received nothing. If turnout is higher in the mailer group, is that because of the mailer's content, or just because the mailer reminded them that an election was happening?

A placebo control solves this. The placebo group receives something similar in form but neutral in content. In a campaign ad study, the placebo might be a public service announcement of the same length. In a mailer experiment, the placebo might be a postcard about an unrelated civic topic. The placebo strips away the form of the treatment so that any remaining difference reflects only the substantive content.

> **Common Mistake: No-Contact Control as Placebo**
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> If your treatment group gets contacted and your control group does not, you are measuring "the effect of being contacted" rather than "the effect of the message." A real placebo holds form constant and varies only the content of interest.

1.10 Survey Length and Respondent Fatigue

Most political experiments are run on survey platforms. That introduces a constraint many researchers underestimate: respondents get tired. As surveys get longer, completion rates drop and response quality degrades. Participants start straight-lining (giving the same answer to every question), skipping items, or speeding through without reading.

> Research on survey fatigue suggests a rough rule of thumb. After about 15 minutes, response quality begins to decline noticeably. By 30 minutes, completion rates can drop below 70% and quality scores fall by half. The same experiment will produce different results depending on

> whether it is placed at minute 5 or minute 25 of a survey.

The practical advice: keep experiments short, place the most important measures early, and randomize the order of modules within a survey so that fatigue affects all conditions equally rather than biasing one in particular.

A related concern is the manipulation check. After delivering a treatment, you can ask participants a question that verifies they actually received and understood it. If most participants in the treatment group cannot answer the manipulation check correctly, a null result is ambiguous: maybe the treatment had no effect, or maybe nobody noticed it. Reporting manipulation check pass rates alongside results lets readers judge for themselves how strong the treatment really was.

1.11 Random Assignment

Random assignment distributes treatment by chance; random sampling selects units by chance. Both matter but for different reasons — random assignment is what licenses causal claims, while random sampling is what licenses generalization to a population.

Everything in this section depends on one assumption: that treatment was assigned by chance. Random assignment is what gives experiments their causal authority. It ensures that on average, the treatment group and the control group are equivalent on every characteristic, including ones we did not measure. If turnout is higher in the treatment group, it is not because the treatment group was younger, more educated, or more politically engaged to begin with. Randomization broke any such correlation by construction.

Random assignment is different from random sampling. Random sampling is about who gets into the study; random assignment is about which condition they end up in. A study can have random assignment without random sampling (most lab experiments) or random sampling without random assignment (most observational surveys). For causal inference, what matters is random assignment.

> **Why Randomization Works**
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> Suppose we want to know whether a campaign ad shifts vote intention. We could compare people who saw the ad to people who did not. But people who saw the ad probably differ in countless ways: they watch more TV, follow more politics, live in different media markets, support different parties to begin with. Any of those differences could explain the result.
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> Random assignment fixes this. By flipping a coin to decide who sees the ad, we guarantee that on average the two groups have the same media habits, the same political interest, the same prior partisanship. The only systematic difference is the ad itself.

In practice, randomization can fail. People drop out (attrition). Some assigned to treatment never receive it (non-compliance). Treatment leaks from treated to untreated units (spillover). Each of these threatens the comparability that randomization was supposed to guarantee, and welldesigned experiments include strategies for handling them: intent-to-treat analysis, bounds, cluster randomization, and so on. But the starting point is always the same: assign treatment by chance, and the rest of inference follows.

The classic illustration of random assignment in political science is the Gerber and Green (2000) get-out-the-vote experiment in New Haven. Researchers took 30,000 registered voters and randomly assigned them to receive a phone call, a piece of direct mail, an in-person canvasser, or nothing at all. Because assignment was random, the four groups were comparable on age, partisanship, prior turnout history, and every other characteristic. When face-to-face canvassing produced a 9.8 percentage point increase in turnout while phone calls produced no effect, those numbers could be interpreted as causal effects of the contact method itself rather than artifacts of who got contacted.

1.12 After Randomization: Attention, Compliance, and Attrition

Random assignment guarantees comparable groups at the moment of assignment. Several things can erode that comparability before you analyze the data.

Attention and data quality. Most online samples — MTurk, Prolific, and the like — contain speeders, satisficers, and outright bots. A short **attention check**, an item that instructs careful readers to give a specific answer, catches respondents who are not really reading. Inattentive participants add noise and can wash out a real effect, so screening them improves your data. The one rule: decide your exclusion criterion in advance and pre-register it. Dropping participants after you have peeked at the results is a back door to p-hacking.

Compliance. Assignment tells people what to do; it cannot make them comply. In a canvassing experiment you might assign 500 people to be contacted, but only 350 answer the door. You now have two different quantities. The **intent-to-treat** (ITT) effect compares everyone *assigned* to treatment against everyone assigned to control, regardless of who complied; it is conservative, honest, and preserves the randomization. The **treatment-on-treated** effect estimates the effect among those who actually complied; it is larger, but compliers are a self-selected group, no longer guaranteed comparable to the controls. Report ITT as your headline — it answers the question a campaign actually faces: if I *send* the canvassers, what happens?

Differential attrition. Some dropout is harmless. If 15 percent of both groups leave, you lose statistical power but keep comparability. The danger is **differential** attrition: if more people leave the treatment group — perhaps because the treatment is unpleasant — then the survivors in treatment differ systematically from the survivors in control, and randomization is broken. Any difference in outcomes could then reflect who dropped out rather than the treatment. You cannot fully fix this after the fact; prevent it by keeping the burden symmetric across arms and tracking who leaves.

1.13 How Many Participants? Statistical Power

Before collecting data, ask whether your study is large enough to detect the effect you expect. **Statistical power** is the probability of detecting a real effect when one exists; it rises with larger effects, larger samples, and less noisy outcomes. Underpowered studies fail in two directions at once. They miss real effects, and — less intuitively — when a small study does reach significance, the estimated effect is almost always inflated, because the result crossed the threshold only when noise happened to help. An honest null from a well-powered study is worth more than an exciting result from an underpowered one. The practical move is to estimate the sample you need in advance, for the smallest effect you would care about, rather than collecting whatever is convenient and hoping.

2 Using AI for This Material

Where AI Helps This Week

- Explaining the distinctions among vignette, conjoint, factorial, and field experiments with worked examples of each.
- Drafting manipulation-check items that confirm participants noticed the treatment without revealing the hypothesis.
- Suggesting balance tests to run on pre-treatment covariates after random assignment.

Where AI Gets This Wrong This Week

- Judging whether a specific protocol clears IRB or ethical review depends on participant risk, deception, and debriefing — context AI does not have.
- Estimating realistic effect sizes for power calculations requires field-specific prior literature, not first-principles guessing.
- It will not spot demand effects baked into your specific stimulus wording, because it cannot read the participant's mind or your stimulus in context.

A prompt that works for this week:

> Draft three manipulation-check questions for a vignette experiment where one condition describes a candidate as a Democrat and the other as a Republican. The questions should verify that respondents noticed the partisan cue without revealing the hypothesis.

A prompt that misleads:

> What sample size do I need to detect the effect of my treatment?

Why it misleads: power calculation requires a realistic effect-size guess grounded in prior studies of comparable interventions; without that anchor, AI will pick a generic $d = 0.2$ or $d = 0.5$ and produce an N that has nothing to do with your actual design.

3 Practice Problems

Conceptual Questions

First, what does random assignment guarantee that random sampling does not? Why is this distinction important for causal inference?

Second, you have a 2x2 factorial design crossing campaign message tone (positive vs. negative) with messenger party (Democrat vs. Republican). Describe the four conditions and the three statistical comparisons a 2x2 design lets you make.

Third, you want to test whether a candidate's military service raises voter support. Identify a clean treatment (what you would vary, X), a sensible control condition, and one behavioral and one attitudinal outcome you could measure (Y). Explain why a no-information control might confound your estimate.

Fourth, explain in plain language what it means to say "the difference between groups is statistically significant at $p < 0.05$." Why is this not the same as saying the difference is large, and how does statistical power enter the picture?

Fifth, a canvassing experiment assigns 1,000 voters to be contacted and 1,000 to control, but only 600 of the treatment group answer the door. Turnout is 40% among everyone assigned to treatment and 34% among controls. What is the intent-to-treat effect? Will the treatment-on-treated effect be larger or smaller, and which would you report?

Sixth, an experiment's treatment is a long, upsetting video and its control is a short, pleasant one. More people quit the survey in the treatment group. Why is this differential attrition more dangerous than dropout that affects both groups equally?

For Further Study

The classic reference on field experiments is Gerber and Green's *Field Experiments: Design, Analysis, and Interpretation*. Hainmueller, Hopkins, and Yamamoto's work on conjoint designs is the standard introduction to that method, and Mutz's *Population-Based Survey Experiments* covers survey-experimental design in depth. For a broad overview of the field, see Druckman, Greene, Kuklinski, and Lupia's *Cambridge Handbook of Experimental Political Science*.